LINEAR ALGEBRA FINAL

07.07.2021

On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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b)

$$\begin{bmatrix} 1 & \alpha & -7 & \alpha \\ 3 & 5 & -9 & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & 6 & 2 \end{bmatrix} = A$$

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These elements axe 0.

$$\begin{bmatrix} 1 & \alpha & -7 & \alpha \\ 0 & 5 & -9 & 1 \\ 0 & 0 & 3 & \alpha^2 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

α and $\beta = 0$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Eliminated upper triangular matrise U.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

a) $A=\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$$

$k=14$
$\alpha=0$
$\gamma=0$

$\alpha^2=0 \Rightarrow \alpha=0$
$\beta=-1$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$m=2$
$k=14$
$\alpha=0$
$\gamma=0$
$\beta=-1$

(3) a) $H^2 = H.H = (I - 2uu^T).(I - 2uu^T)$

$= I^2 - I.2uu^T - 2uu^T.I + 4uu^T.uu^T$

$= I(I - 2uu^T - 2uu^T) + 4u.\|u\|^2.u^T$

$= I - 2uu^T - 2uu^T + 4u.\|u\|^2.u^T$

$\|u\|^2 = u^T u$

$\|u\| = \pm\sqrt{u^T u}$

$\|u\| = 1$

So, $H^2 = I - 4uu^T + 4uu^T$

$H^2 = I - 4(1) + 4(1)$

$\boxed{H^2 = I}$

b) $HH^T = (I - 2uu^T)(I - 2uu^T)^T$

$= (I - 2uu^T)(I^T - (2uu^T)^T)$

$= (I - 2uu^T)(I - 2(u^T)^T u^T)$

$= (I - 2uu^T).(I - 2uu^T)$

$= I^2 - 2uu^T - 2uu^T + 4uu^T.uu^T$

$= I - 4uu^T + 4u\|u\|^2 u^T$

$= I - 4uu^T + 4uu^T$

$= I - 4(1) + 4(1)$

$H.H^T = I$

$H^T H = I$

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

So, $HH^T = H^T H = I$

So, "H" is an Orthogonal matrix.

c) $Av = \lambda v$

$Hu = (I - 2uu^T).u$

$= (Iu - 2uu^Tu)$

$= u - 2u\|u\|^2$ As I is identity matrix

$Iu = 1u = u$ and $\|u\| = 1 = \pm\sqrt{u^Tu}$

hence, $u^Tu = \|u\|^2 = 1$

$Hu = u - 2u.1 = -u$

$Hu = -u = (-1)u$

$Hu = \lambda u$

$\lambda u = -1$

So, u is an eigen vector of H and corresponding eigen value for u is $= \lambda = -1$

d) $H = (I - 2uu^T)$

$H^T = (I - 2uu^T)^T$

$= I^T - 2(uu^T)^T$

$= I - 2(u^T)^T.u^T$

$= I - 2uu^T = H$

So, H is a symmetric matrix and there is a theorem which states that,

$H = Q\Lambda Q^T$

$= Q\Lambda Q^{-1}$

$= (Q^{-1})^{-1}\Lambda Q^{-1}$

$= P^{-1}\Lambda P$

- Q is an orthogonal matrix
- $QQ^T = I \rightarrow Q^T = Q^{-1}$
- Λ = Diagonal matrix consists of eigen value of H.

So, H is diagonizable matrix.